$$\widetilde{D}\left[\hat{\sigma}_z,\left[\hat{\sigma}_z,\hat{\rho}_{\mathcal{S}}(t)\right]\right]=\widetilde{D}\left(\frac{1}{2}\hat{\rho}_{\mathcal{S}}(t)-2\hat{\sigma}_z\hat{\rho}_{\mathcal{S}}(t)\hat{\sigma}_z\right) \tag{5.146}$$